



Statistics of Financial Markets

Chapter 11: Machine Learning in Finance



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Outline

Motivation

LASSO Regression

Random Forest

Model Validation

Applications in Finance

Conclusions

References

Motivation

- ▣ The rise of machine learning in quantitative finance
- ▣ From traditional econometrics to data-driven approaches
- ▣ Bias-variance tradeoff and the curse of dimensionality

LASSO Regression

- L_1 -penalized regression: $\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$
- Sparsity and automatic variable selection
- Cross-validation for tuning λ
- Comparison with Ridge regression and Elastic Net
- Applications: factor selection in asset pricing

Random Forest

- Decision trees as base learners
- Bagging and random feature subsampling
- Out-of-bag error estimation
- Variable importance measures
- Advantages: non-linearity, robustness to outliers

Model Validation

- Train-test split and k -fold cross-validation
- Time series cross-validation (expanding window, rolling window)
- Overfitting detection and regularization
- Performance metrics: RMSE, MAE, R^2 , accuracy, F1-score

Applications in Finance

- ▣ Stock return prediction and portfolio construction
- ▣ Credit risk modeling with ML methods
- ▣ Fraud detection and anomaly detection
- ▣ Python implementation using `scikit-learn`

 [SFM_ch11_machine_learning](#)

Conclusions

- ▣ Summary of LASSO and Random Forest methodologies
- ▣ When to use ML vs. traditional econometric models
- ▣ Interpretability vs. predictive performance tradeoff

References

- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso.
- Breiman, L. (2001). Random forests.
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